

Name _____

Date Feb 27, 2025

1. Differentiate the following functions. Express answers in radical form.

a) $f(x) = -\frac{1}{3}x^3 + 4x^2 - 8$

$$f'(x) = -\frac{1}{3} \frac{d}{dx}(x^3) + 4 \frac{d}{dx}(x^2) - \frac{d}{dx}(8)$$

$$= -x^2 + 8x$$

b) $y = 9x^{-2} + 4\sqrt{x}$

$$f'(x) = 9 \frac{d}{dx}(x^{-2}) + 4 \frac{d}{dx}(x^{1/2})$$

$$= -\frac{18}{x^3} + \frac{2}{\sqrt{x}}$$

(3)

2. Determine the coordinates of the point(s) on the graph of $y = 2x^5 - 160x$ where the tangent is horizontal.

$$y = 2x^5 - 160x$$

$$f'(x) = 2 \frac{d}{dx}(x^5) - 160 \frac{d}{dx}(x)$$

$$= 10x^4 - 160 \leftarrow m$$

$$0 = 10x^4 - 160$$

$$\frac{160}{10} = \frac{10x^4}{10}$$

$$y = x^2$$

$$m = 0$$

$$x = \pm 2 \leftarrow (\text{and } 2i \text{ and } -2i)$$

$$f(2) = 2(2)^5 - 160(2)$$

$$= -256$$

$$f(-2) = 2(-2)^5 - 160(-2)$$

$$= 256 \quad |P(2, -256) \quad P(-2, 256)|$$

(3)

3. A football is kicked up into the air. Its height, h , above the ground, in metres, at t seconds can be modelled by $h(t) = 18t - 4.9t^2$. Determine $h'(3)$, and describe what it represents.

$$f'(x) = 18 \frac{d}{dx}(x) - 4.9 \frac{d}{dx}(x^2)$$

$$= 18 - 9.8x$$

$$f'(3) = 18 - 9.8(3)$$

$$= -11.4 \text{ m/s}$$

$$h'(3) = -11.4 \text{ m/s}$$

represents that at 3 seconds, the football is at a velocity of

$$-11.4 \text{ m/s}$$

(2)

4. Determine the derivative of $f(x) = \sqrt{2x-5}$ using first principles.

$$P(x, \sqrt{2x-5}) \quad Q(x+h, \sqrt{2(x+h)-5})$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{2(x+h)-5} - \sqrt{2x-5}}{h} \times \frac{\sqrt{2(x+h)-5} + \sqrt{2x-5}}{\sqrt{2(x+h)-5} + \sqrt{2x-5}}$$

$$= \lim_{h \rightarrow 0} \frac{2x+2h-5-2x+5}{h(\sqrt{2(x+h)-5} + \sqrt{2x-5})} = \frac{2}{(\sqrt{2(0+x)-5} + \sqrt{2x-5})}$$

$$= \frac{2}{\sqrt{2x-5} + \sqrt{2x-5}}$$

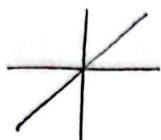
$$= \frac{1}{\sqrt{2x-5}}$$

$$f'(x) = \frac{1}{\sqrt{2x-5}} \quad (2)$$

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5. Discuss the continuity and differentiability of each function. If not differentiable, state the type of non-differentiability.

a) $f(x) = x|x|$



① $f(x)$ is cont. $\forall x \in \mathbb{R}$

② $f(x) = \begin{cases} x > 0, & x(x) \\ x < 0, & x(-x) \end{cases}$

for $x > 0$, $f'(x) = \frac{d}{dx}(x^2)$
 $= 2x$

for $x < 0$, $f'(x) = -\frac{d}{dx}(x^2)$
 $= -2x$

$\lim_{x \rightarrow 0^+} f'(x) = 0$ $\lim_{x \rightarrow 0^-} f'(x) = 0$

$\therefore$ the $f(x)$ is differentiable $\forall x \in \mathbb{R}$,

6. Determine the equations of the tangents to the curve $y = -3x^2 - 5$ that pass through the point $(-1, 4)$

$y = -3x^2 - 5$

$f'(x) = -3 \frac{d}{dx}(x^2) - \frac{d}{dx}(5)$
 $= -6x \leftarrow m$

$\frac{y_2 - y_1}{x_2 - x_1} = m$

$P_1 = (-1, 4)$

$P_2 = (x, -3x^2 - 5)$

$\frac{-3x^2 - 5 - 4}{x - (-1)} = -6x$

$-3x^2 - 9 = -6x^2 - 6x$

$3x^2 + 6x - 9 = 0$

$x = 1, -3$

$f'(1) = -6(1)$
 $= -6$

$f'(-3) = -6(-3)$
 $= 18$

$y = -6(-1) + b$

$b = -2$

$y = -6x + 2$

$y = 18(-1) + b$

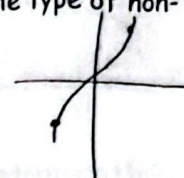
$b = 22$

$y = 18x + 22$

$\boxed{\begin{matrix} y = -6x + 2 \\ y = 18x + 22 \end{matrix}}$

Ans 8.5

b) $g(x) = 10x^{\frac{1}{5}}$



① $f(x)$ is cont $\forall x \in \mathbb{R}$

② $f(x) = 10x^{\frac{1}{5}}$

$f'(x) = 10 \frac{d}{dx}(x^{\frac{1}{5}})$

$= 2x^{-\frac{4}{5}}$

$= \frac{2}{\sqrt[5]{x^4}} \leftarrow \text{DNE @ } x = 0$

$\lim_{x \rightarrow 0^+} f'(x) = +\infty$ $\lim_{x \rightarrow 0^-} f'(x) = +\infty$

matching $f'(x)$ limits

$\therefore f(x)$ is non-differentiable

at $x = 0$ due to Vertical Tangent

8.5